



1. What is the value of $\frac{2^2+2^0}{2^2+2^4}$?

- (A) $\frac{1}{4}$ (B) $\frac{3}{10}$ (C) $\frac{5}{16}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

2. Elise buys 1 raffle ticket for her school's fundraiser. To determine the winner of the raffle, a ticket is randomly selected among all raffle tickets purchased, and Elise wins the raffle if her ticket is selected. If Elise had instead bought 100 raffle tickets, she would be 82 times more likely to win the raffle. What are Elise's chances of winning the raffle if she buys only 1 raffle ticket?

- (A) $\frac{1}{492}$ (B) $\frac{1}{451}$ (C) $\frac{1}{428}$ (D) $\frac{1}{420}$ (E) $\frac{1}{361}$

3. Saul has a sock drawer with socks that are either black or white, and either long or short. One day, while reorganizing his drawer, Saul found that:

- His white socks and his black socks are in ratio 4 : 5;
- His long white socks and his long black socks are in ratio 2 : 3;
- His short white socks and his short black socks are in ratio 4 : 3.

What is the ratio of Saul's long socks to his short socks?

- (A) 3 : 1 (B) 5 : 4 (C) 13 : 8 (D) 18 : 5 (E) 20 : 7

4. Aarav, Brian, and Corey are standing in a line on an empty field so that Aarav is 133 meters away from Brian and 397 meters away from Corey. Brian and Corey are told to clap immediately after hearing Aarav clap once. After Aarav claps once, he hears two claps from Brian and Corey, one 1.6 seconds after the other. Which of the following is closest to the speed at which sound traveled during this experiment, in meters per second?

- (A) 330 (B) 335 (C) 340 (D) 345 (E) 350

5. Rosemary has a peculiar analog clock that chimes when its hour and minute hands are perpendicular to each other. Her clock currently displays the time 7:00 PM. The next

time her clock chimes occurs between x and $(x + 1)$ minutes from now, where x is a positive integer. What is x ?

- (A) 5 (B) 6 (C) 16 (D) 21 (E) 22

6. What is the remainder when $(2!)^{(3!)}(4!)^{(5!)}\dots(2024!)$ is divided by 10?

- (A) 0 (B) 2 (C) 4 (D) 6 (E) 8

7. Let x and y be positive integers such that y is a factor of 143^x , and y has exactly $x^2 + 143$ positive factors. What is the sum of the digits of the sum of all possible values of x ?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

8. Let P be the (unique) monic polynomial of minimum degree such that P has integer coefficients and $P(1 + \sqrt[3]{5}) = 0$. What is the value of $P(1 - \sqrt[3]{5})$?

- (A) -10 (B) -5 (C) 0 (D) 5 (E) 10

9. Ana has a four digit number, and assigns each digit to one of the variables a , b , c , or d . She notices that no matter how she assigns the digits to a , b , c , and d , the value of $ab + bc + cd + da$ is always even. Given that Ana's friend Bailey now selects a four digit number at random, what is the probability his number also satisfies this property?

- (A) $\frac{1}{2}$ (B) $\frac{7}{12}$ (C) $\frac{5}{8}$ (D) $\frac{13}{18}$ (E) $\frac{3}{4}$

10. Suppose z is a complex number and x and y are nonzero reals such that $|z| = |z + x| = |z + yi| = 1$. What is $x^2 + y^2$?

- (A) $\frac{\sqrt{2}}{2}$ (B) $\sqrt{2}$ (C) 2 (D) $2\sqrt{2}$ (E) 4

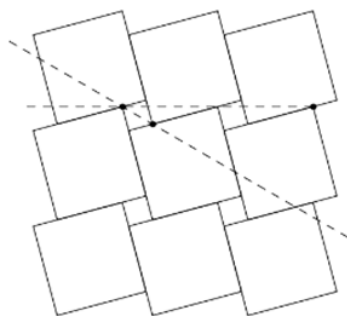
11. Convex quadrilateral $ABCD$ has area 100. Point E lies on AB , and points F and G are the midpoints of CE and DE , respectively. If the area of quadrilateral $AGFB$ is 40, what is the area of $\triangle EFG$?

(A) 5 (B) 10 (C) 15 (D) 20 (E) 25

12. Let a be a real number and $a \neq 1$. What is the minimum possible integral value of $\left|a + \frac{10}{a-1}\right|$?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

13. The side length of the larger tiles is denoted as x , and the side length of the smaller tiles is denoted as y . The dotted lines form a 30° angle. What is the ratio x/y ?



(A) 2 (B) $2\sqrt{3}$ (C) $2 + \sqrt{3}$ (D) $3\sqrt{2}$ (E) $3 + \sqrt{2}$

14. Let the sequence (a_n) satisfy $a_1 = 5$ and $a_1 + a_2 + \cdots + a_n = n^2 a_n$ for all integers $n > 1$. What is a_{25} ?

(A) $\frac{1}{625}$ (B) $\frac{1}{120}$ (C) $\frac{1}{65}$ (D) 625 (E) 1625

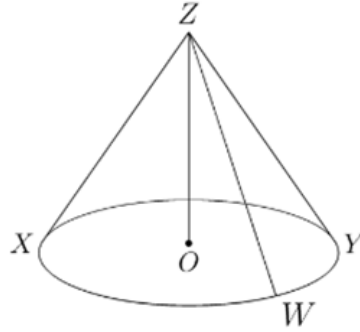
15. 12 distinct mugs are each painted either red, white, blue, or yellow. Let n be the number of ways to paint these mugs such that an odd number of them are painted blue and an even number of them are painted yellow. What is the remainder when n is divided by 5?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

16. On rectangle $ABCD$ with area 1, the midpoints of AB and BC are M and N , respectively. Point X is chosen on segment AD and the intersections of MX and NX with AC are points Y and Z , respectively. If $AC = 4YZ$, what is the area of triangle XYZ ?

- (A) $\frac{1}{18}$ (B) $\frac{1}{16}$ (C) $\frac{1}{12}$ (D) $\frac{1}{9}$ (E) $\frac{1}{8}$

17. In the given cone, the diameter XY of the base circle is 2 units. Point Z is positioned directly above the center O of the base circle, such that $ZO = \sqrt{2}$. A point W lies on the circle such that the ratio of the arc length XW to arc length WY is $2 : 1$. What is the shortest distance between X and the line ZW ?



- (A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\sqrt{3}$ (D) 2 (E) $2\sqrt{2}$

18. Six distinct positive integers from 1 to 35 are chosen uniformly and independently at random. Given that 1 is one of these six integers, and it is possible to place these integers on the 6 vertices of a regular hexagon such that the 3 pairs of opposite vertices have the same sum, what is the expected value of this common sum?

- (A) $\frac{51}{2}$ (B) $\frac{53}{2}$ (C) $\frac{55}{2}$ (D) $\frac{57}{2}$ (E) $\frac{59}{2}$

19. The sum of all real numbers a for which $\sqrt[3]{2 + \sqrt{a}} + \sqrt[3]{2 - \sqrt{a}}$ is an integer can be written as $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $p + q$?

- (A) 6 (B) 127 (C) 135 (D) 262 (E) 289

20. A line ℓ passing through vertex A of equilateral triangle ABC cuts through the triangle. The distances from B and C to line ℓ are 10 and 7, respectively. What is BC^2 ?

- (A) 282 (B) 292 (C) 302 (D) 312 (E) 322

21. Let $f(x) = x^3 + 11x - 5$ have roots a, b, c . What is $f(a+1)f(b+1)f(c+1)$?

(A) 1716 (B) 3402 (C) 6885 (D) 7155 (E) 9416

22. The volume of the region in the positive xyz -space defined by the equation $\lfloor x + y \rfloor + \lfloor y + z \rfloor + \lfloor z + x \rfloor = 21$ can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

(A) 17 (B) 59 (C) 70 (D) 89 (E) 125

23. What is the sum of the digits of the smallest positive integer n such that $n^n + 2^n$ is divisible by 1000?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

24. Given a positive integer n , let x_n denote the number of ordered pairs of integers (a, b) such that $a^n + b^n$ is divisible by 109, where $0 \leq a, b \leq 108$. What is $x_1 + x_2 + \cdots + x_{108}$?

(A) 34884 (B) 34992 (C) 35100 (D) 35208 (E) 35316

25. An ant sits at $(0, 0)$ on the Cartesian plane. Each minute, it marches either one unit up or one unit right. Suppose there are n distinct paths the ant can take to $(10, 10)$ that intersect the line $x = y$ at exactly 5 points (not including $(0, 0)$ and $(10, 10)$). What is the sum of the digits of n ?

(A) 15 (B) 18 (C) 21 (D) 24 (E) 27

Time limit: 75 minutes